

Data-Driven Results for Work Packages 10 and 11

Digital twins per building with edge integration; feedback control with continuous learning for self-optimizing buildings

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Scope and evidential basis

This report summarizes the measured results obtained under work packages 10 and 11. All experiments were run on the BOPTEST framework, an open Modelica-based benchmark for building control, using four single-zone test cases: the air-based source case `bestest_air` and three hydronic cases spanning residential and commercial scale. Every number below is traceable to a committed result file; the provenance column of each table names it.

Two objectives within these work packages were not reached in the reporting period. They are reported as such under *Objectives not reached* below rather than presented as partial successes, because both are prerequisites for the 2027 activities and the schedule depends on an accurate status.

A peer-reviewed output covering the core of both work packages was submitted to *Energies* (MDPI) on 9 August 2026, manuscript `energies-4523055`, and is under editorial assessment.

Work package 10: digital twins per building and edge integration

10.1 Twin architectures and predictive validity

Two surrogate architectures were developed for the source building and validated against multi-step BOPTEST rollouts rather than one-step residuals, since a twin used for control is evaluated on trajectories, not on single transitions.

Twin	Architecture	24 h rollout RMSE (°C)
BB (hourly)	compact black-box, 8482 parameters, multi-horizon rollout objective	1.557
BB (15-min)	same architecture, matched 900 s corpus	0.876
GB (raw)	grey-box RC balance with neural residual heat-flow head, uncalibrated	1.466
GB (calibrated)	same, after Stage A/B/C inverse identification	0.644

RMSE is the multi-step rollout root-mean-square error of zone temperature over a 24-hour horizon. Provenance for all four rows: `block1_corpus_matched_comparison.csv`.

The improvement from the hourly black-box baseline to the calibrated twin is 0.913 °C. A corpus-matched ablation attributes 0.681 °C (74.6%) to corpus temporal resolution and 0.232 °C (25.4%) to the staged calibration itself. An alternative decomposition through the uncalibrated grey-box backbone assigns 90.1% to calibration; the two differ because corpus and architecture interact non-additively, so the calibration contribution is reported as a range rather than a single number.

10.2 Physical identifiability of the twin

The grey-box twin carries one physically interpretable parameter, the zone thermal capacitance C_{zon} , recovered by staged inverse identification from telemetry.

- Identified value $C_{\text{zon}} = 4.413 \times 10^5 \text{ J/K}$, a +5.1% update over the physical prior and within one standard error of the Laplace/Fisher interval.
- Physically equivalent to approximately 439 kg of air, the expected order of magnitude for a 48 m² zone.
- Directional validity: sweeping the supply-temperature command at fixed state gives the physically correct response sign in 100% of 400 sampled states.
- Temperature-alignment stage reduced alignment RMSE from 0.374 °C to 0.232 °C, a 38.0% improvement (block1_surrogate_final_metrics.csv).

10.3 Twins for each building: recalibration across the building family

The objective of a twin *per building* was tested by porting the identification pipeline to three further test cases that differ in heat source, distribution path and actuator interface. The pipeline transferred to all three.

Building	Zone temperature			HVAC power		ρ_C
	raw RMSE (°C)	calibrated (°C)	gain (%)	raw MAE (W)	calibrated (W)	
Residential, heat pump (192 m ²)	1.421	0.565	60.2	2921	1767	1.892
Residential, boiler (48 m ²)	2.666	0.335	87.4	784	85	1.954
Commercial, district heating (8500 m ²)	1.952	0.238	87.8	—	—	1.909

ρ_C is the re-identified zone capacitance as a ratio to the source building. Provenance: block3_hydraulic_family_n2_summary.csv, block3_transfer_matrix.csv.

Two results are worth isolating. First, recalibration reduced rollout error by 60.2–87.8 % on every target, so the identification pipeline is a portable asset rather than a one-building artifact. Second, the re-identified capacitance clusters at $\rho_C = 1.918 \pm 0.032$ across an order-of-magnitude change in floor area (48–8500 m²). This is reported as an effective single-zone RC signature of the hydronic family and not as a claimed physical invariant: it rests on three cases.

10.4 Runtime characteristics relevant to edge deployment

Throughput was benchmarked against the live BOPTEST runtime under an identical 900 s control period, on a single CPU thread with no GPU.

Backend	Env. steps/s	Mean step (ms)	Speed-up
BOPTEST RTE (HTTP)	21.0	15.33	1×
BB surrogate	4626	0.19	220.2×
GB calibrated surrogate	—	—	114.2×
Role-separated hybrid	—	—	85.0×

Provenance: speed_benchmark_table.csv. Device cpu, torch_num_threads = 1. At hybrid throughput a 5×10^6 -step policy-optimization run costs about 47 minutes of environment time against roughly 66 hours on the live runtime.

These figures characterize the computational footprint of the twin: an 8482-parameter model executing a control step in 0.19 ms on one commodity CPU thread. That is the quantitative basis on which edge deployment becomes feasible. It is not itself an edge deployment; see *Objectives not reached* below.

Work package 11: feedback control with continuous learning

11.1 Closed-loop control deployed on the live runtime

Feedback control was deployed against the live BOPTEST Run-Time Environment, a Dockerized Modelica emulator with embedded baseline loops exposed over a REST API. At each 900 s step the agent reads measurements and disturbance forecasts, and overwrites a single control input, the supply-air temperature setpoint, leaving the remaining embedded loops at baseline. Controllers are scored with the duration-and-severity maintenance score $m_s = r_{\text{time}} + r_{\text{sev}}$, where $m_s > 1$ marks an unusable controller.

Training backend	m_s (seed $42 \pm$ seed s.d.)		Comfort violation (%)	
	peak	typical	peak	typical
BB hourly	0.073 ± 0.019	0.095 ± 0.047	1.5	4.4
BB matched-resolution	1.142 ± 0.000	1.211 ± 0.000	85.6	91.4
GB calibrated	1.046 ± 0.022	1.102 ± 0.026	77.1	82.4
Role-separated hybrid	0.087 ± 0.026	0.041 ± 0.024	4.7	2.4
BOPTEST built-in PI (12-month)	0.910		63.6	

Dispersions are standard deviations over seeds {42, 43, 44}. Provenance: `block2_fidelity_utility_scatter.csv`, `block2_thermostatic_seed_band.csv`.

The principal scientific result of the period sits in this table. The most accurate twin (0.644°C) trains a controller that fails on every seed, while the least accurate (1.557°C) trains a usable one; making the black-box twin more accurate by refining its resolution moves it from the usable regime into the failing one. Predictive accuracy is therefore not a valid criterion for selecting a twin as a training environment. A rescaling control that holds scale-free response-surface roughness fixed while varying only the integration step reproduces the collapse, isolating the per-step increment magnitude as the operative property. Measured relative roughness: 0.169 for the usable hourly twin against $9.4\times$ and $7.9\times$ that for the two failing backends (`block2_mechanism_surface_sharpness.csv`).

11.2 Role separation as the deployed control architecture

The architecture that resolves the conflict assigns the two conflicting requirements to two models: the black-box twin supplies the rollout dynamics, and the frozen calibrated twin acts only as a per-step model-disagreement reward censor, never entering the rollout or the policy loss. Averaged over three seeds this backend leads the best single-model alternative on both evaluation windows (0.060 against 0.072 on the peak window, 0.014 against 0.046 on the typical window), at $85\times$ live throughput. Mean twin disagreement across hybrid traces is 0.92°C and 682 W on the peak window and 1.01°C and 735 W on the typical window (`hybrid_disagreement_summary.csv`).

11.3 Controller families and the observation interface

- **Hierarchical DRL.** A seasonal hierarchy routing to winter and summer setpoint specialists degrades as the disagreement censor is strengthened: peak comfort violation rises from $6.3 \pm 1.7\%$ at $\lambda_T = 0$ to $24.4 \pm 3.8\%$ at $\lambda_T = 0.10$, the setting optimal for the thermostatic family. The sweep is replicated over seeds {42, 43, 44} at all four settings: the endpoint gap is 10.0 (peak) and 5.8 (typical) pooled seed standard deviations, although the $0.05 \rightarrow 0.10$ increment sits within one pooled s.d. so the two strongest weights are not separable. The regularization strength is controller-family specific and cannot be quoted without naming its family (`block2_hdr1_lambda_sweep_seed_band.csv`).
- **Multi-objective RL and the observation interface.** Widening the observation from a 5-dimensional instantaneous vector to a 17-dimensional forecast-augmented one changes the controller from unus-

able to usable: RMSE falls from 4.96 °C to 0.72 °C, comfort violation from 74.5% to 4.9%, and m_s from 1.046 to 0.099. The binding constraint is observation geometry, not the reward scalarization (block2_morl_comparison_summary.csv).

- **Multi-objective stability.** Over five fixed seeds the neutral preference point attains $m_s = 0.187 \pm 0.078$ (coefficient of variation 0.42, range 0.103 to 0.310). The mean is well inside the usable band but the spread fails a pre-specified stability test, so the preference-conditioned controller is not yet deployment-stable (morl_canonical_seedfix_yearly_summary.csv).

11.4 Evidence bearing on the continuous-learning objective

Three results from the period speak directly to whether a frozen policy can be deployed across a building stock, which is the premise the continuous-learning objective is meant to relax.

1. **Frozen-policy transfer fails on two of three target buildings.** Both residential cases fall short of the pre-specified $1.25 \times$ -PI comfort threshold ($m_s = 0.665$ against a threshold of 0.579; $m_s = 0.976$ against 0.938) while saving 7.3% and 5.8% energy respectively. The commercial case passes the comfort threshold ($m_s = 0.431$ against 0.785) but consumes 35.3% more energy than the PI baseline, so it is a threshold pass and not a deployment-ready result.
2. **The transferable asset is the pipeline, not the policy.** Against the same three targets the identification pipeline succeeded in every case (Section 10.3). Transfer is therefore component-level: identified physics ports across buildings, the frozen control law does not.
3. **Warm-starting from the physical twin is counterproductive.** Pretraining on the calibrated twin and then fine-tuning on the live runtime raises m_s by a factor of two to three on both windows against an otherwise identical from-scratch fine-tune. Adaptation strategies for the continuous-learning objective should not assume that twin-pretrained weights are a useful starting point.

Taken together these quantify the need that work package 11 addresses: local adaptation is required rather than optional, and the adaptation must act on the identified model rather than on transferred policy weights.

Objectives not reached in the reporting period

Edge computing infrastructure integration (work package 10). Not delivered. All experiments ran on a laptop workstation (6-core CPU, 32 GB RAM) with the emulator in a Docker container. No deployment to edge hardware was performed and no edge orchestration was integrated. What the period delivered instead is the quantitative precondition: a twin of 8482 parameters executing a control step in 0.19 ms on a single CPU thread, which establishes that the computational envelope is compatible with edge-class hardware. Integration remains to be carried out.

Continuous online learning (work package 11). Partly delivered. Feedback control is deployed and evaluated in closed loop, and offline adaptation stages exist within the multi-objective pipeline, including a live-runtime fine-tuning stage. What has not been deployed is a continuous online learning loop in which a controller adapts during operation. The transfer results of Section 11.4 establish the requirement quantitatively and identify the correct target for adaptation, but the adaptation mechanism itself is not yet implemented.

Deliverable	Status	Evidence
Digital twins developed for each building	Delivered	Two architectures validated on the source case; identification pipeline ported to three further buildings with 60.2–87.8 % rollout-error reduction on each
Integration with edge computing infrastructure	Not delivered	Computational envelope established (0.19 ms per step, single CPU thread, 8482 parameters); no edge deployment performed
Feedback control strategies deployed	Delivered	Closed-loop control on the live runtime across three controller families; best configuration reaches $m_s = 0.041$ with comfort violation below 5% at $85\times$ live throughput
Continuous learning for self-optimization	Partly delivered	Offline and live fine-tuning stages implemented; transfer study quantifies the need for local adaptation; no online learning loop in operation

Summary of status against the stated deliverables

Recommended priorities for the next period

1. Port the black-box twin to edge-class hardware and measure step latency, memory footprint and thermal behaviour under sustained operation. The single-thread CPU benchmark suggests this is tractable and should be confirmed empirically rather than assumed.
2. Implement local adaptation acting on the identified model, since Section 11.4 shows that policy-weight transfer is the wrong mechanism and model recalibration is the one that ports.
3. Extend seed replication to the remaining controller family. The hierarchical censor-weight sweep is now replicated over three seeds at all four settings, closing what was the weakest link in the controller-family findings; the preference-conditioned MORL controller still fails its pre-specified stability test and is the next target.
4. Add a stronger control baseline. Present comparisons are against the emulator’s built-in PI, which is a weak reference; a tuned rule-based or model-predictive baseline on the same twin would put the reported gains on firmer ground.

All quantities in this report are reproducible from committed result files in the project repository. Figures and tables of the submitted manuscript are linked to their source artifacts through provenance maps in the Supplementary Materials.